

1 — Groups Recap + Euler φ

Group: closure, identity e (unique, $a e e = a$), associativity, inverses (unique). $(\mathbb{Z}, +)$, $(\mathbb{Z}_n, +)$ groups; $\mathbb{Z}_n^*$ group $\iff n$ prime. $|G| = \# \text{elem}$; $\text{ord}(g) = \text{smallest } n > 0 \text{ with } g^n = e$. **Lagrange:** $\text{ord}(g) \mid |G|$, $g^{|G|} = e$. **Cyclic:** $G = \langle g \rangle \iff \text{ord}(g) = |G|$ (one generator makes all).
Worked: $|\mathbb{Z}_8^+| = 8$, $\text{ord}[1] = 8$, $\text{ord}[2] = 4$, $\text{ord}[7] = 8$; gens $[1], [3], [5], [7]$.
 $\varphi(n) = \#\{1 \leq k \leq n : \gcd(k, n) = 1\}$. **Formulas:** p prime $\varphi(p) = p-1$; $\varphi(p^k) = p^k - p^{k-1}$; coprime $\varphi(mn) = \varphi(m)\varphi(n)$.
Worked: $\varphi(16) = 8$ ($1, 3, 5, 7, 9, 11, 13, 15$); $\varphi(17) = 16$.
 $\varphi(14) = \varphi(2)\varphi(7) = (2-1)(7-1) = 1 \cdot 6 = 6$
 $\varphi(9) = 9 - 3 = 6$ (prime power)

Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ if $\gcd(a, n) = 1$ (= Lagrange on units mod n , $|G| = \varphi(n)$). FLT = prime case ($\varphi(p) = p-1$) = Lagrange on $\mathbb{Z}_p^*$. **Use:** prime $\rightarrow$ FLT (smaller exponent); composite $\rightarrow$ Euler. **Idea:** reduce exponent mod $\varphi(n)$.

Worked $3^{347} \bmod 14$: Given $\gcd(3, 14) = 1$, $\varphi(14) = 6$.
Step 1 (reduce exponent): $347 = 6 \cdot 57 + 5$ ($6 \cdot 57 = 342$).
Step 2 (evaluate): $3^5 = 243$;
 $243 - 17 \cdot 14 = 243 - 238 = 5 \Rightarrow [5]$. **Check:** $\text{pow} = 5$.

2 — Crypto

Caesar: $c \equiv m + s \pmod{26}$, $m \equiv c - s$. **Worked** BY ($B=1, Y=24$), $s=20$: $1+20=21=Y$, $24+20=44-26=18=S \Rightarrow$ **VS**.
Power: $c \equiv m^b \pmod{p}$; $m \equiv c^d$, $bd \equiv 1 \pmod{p-1}$. **Worked** $p=29, b=5$: $B: 1^5=1=B$.
Y Step 1 ($24 \equiv -5$): $(-5)^2 = 25$; **Step 2:** $25^2 = 625$; $625 - 21 \cdot 29 = 625 - 609 = 16$.
Y Step 3: $16 \cdot (-5) = -80$;
 $-80 + 3 \cdot 29 = -80 + 87 = 7 \Rightarrow$ **H**. $d: 5d \equiv 1 \pmod{28} \Rightarrow d=17$ ($5 \cdot 17 = 85 = 3 \cdot 28 + 1$).
Correctness $\equiv$ bijective: $D(E(x)) = x$, $E(D(y)) = y$. Caesar always bijective (shift). Power bijective $\iff \gcd(b, p-1) = 1$ (else many-to-one). Many-to-one: ≥ 2 symbols map together; surjective: covers whole target.
Check decrypt: VS-20: $21-20=1=B$, $18-20=-2+26=24=Y$. Power: $7^{17} \bmod 29 = 24=Y$ **Check OK**.

3 — Complex Numbers

Rect $z = a + bi$ ($\text{Re}=a$, $\text{Im}=b$), plane vector (a, b) . Polar $r(\cos \theta + i \sin \theta)$, $r = |z| = \sqrt{a^2 + b^2}$. $|z|^2 = z\bar{z}$, $\bar{z} = a - bi$.
 $+$, $-$ = vector + (parallelogram). Mult: $(a+bi)(c+di) = (ac-bd) + (ad+bc)i$.
Worked $3+4i$: $|z|=5$, $z^{-1} = (3-4i)/25 = 0.12 - 0.16i$.
Worked $(1+6i)/(1-6i)$: $(1+6i)^2/37 = (-35+12i)/37$.
 $\approx -0.9459 + 0.3243i$ **Check** $(37 \cdot (-0.9459) = -35$, $37 \cdot 0.3243 = 12)$.

Euler: $e^{i\theta} = \cos \theta + i \sin \theta$; **DeMoivre:** $(re^{i\theta})^n = r^n (\cos n\theta + i \sin n\theta)$.
 n -th roots: $\sqrt[n]{r}(\cos(\theta+2k\pi)/n + i \sin(\theta+2k\pi)/n)$, $k=0..n-1$.
Div: $z_1/z_2 = z_1\bar{z}_2/|z_2|^2$; **Inverse:** $z^{-1} = \bar{z}/|z|^2$.
Polar mult: $r_1 r_2 e^{i(\theta_1+\theta_2)}$;
 $e^{i\pi/3} \cdot e^{i\pi/6} = e^{i\pi/2} = i$.

Worked $8e^{i\pi/3}$: $8(\frac{1}{2} + i\frac{\sqrt{3}}{2}) = 4 + 4\sqrt{3}i \approx 4 + 6.9282i$.
Worked $11e^{3\pi i/2}$: $11(0-i) = -11i$.
Cube roots of $e^{i\pi/2}$: angles $\pi/6, 5\pi/6, 3\pi/2 \Rightarrow \sqrt{3}/2 + i/2, -\sqrt{3}/2 + i/2, -i$.
Square roots of $e^{i\pi/4}$: angles $\pi/8, 9\pi/8$:
 $\cos \frac{\pi}{8} = \sqrt{2+\sqrt{2}}/2 \approx 0.9239$,
 $\sin \frac{\pi}{8} = \sqrt{2-\sqrt{2}}/2 \approx 0.3827$.
Rect $\rightarrow$ polar: $2+2i$: $r = \sqrt{8} = 2\sqrt{2}$,

$\theta = \pi/4 \Rightarrow 2\sqrt{2}e^{i\pi/4}$ (**Check** $2\sqrt{2}(\sqrt{2}/2 + i\sqrt{2}/2) = 2+2i$).
Geometry: $\times e^{i\theta}$ rotates by θ (lengths mult); $\bar{z}$ mirrors across real axis; unit circle $\{|z|=1\}$ subgroup (closed $|zw|=1$, inv $\bar{z}$). **Plot recipe:** mark Re on x , Im on $y \rightarrow$ arrow; add tip-to-tail; mult adds angles, mults lengths.
Groups: $(\mathbb{C}, +)$ $e=0$ inv $-z$; $(\mathbb{C}^*, \cdot)$ $e=1$ inv $\bar{z}/|z|^2$; $C_4^* = \{\pm 1, \pm i\}$ gen i ; C_8^* gen $(1+i)/\sqrt{2}$; $\{1, -1\}$ group (gen -1); $\{i, -i\}$ **not** ($i \cdot i = -1$ missing).
Mult prop $|z_1 z_2| = |z_1| |z_2|$: $z_1 = (1+i)/\sqrt{2}$, $z_2 = (1-i)/\sqrt{2}$: $1 \cdot 1 = |1| = 1$ **Check OK**.

4 — Roots of Unity

Powers $\omega^0.. \omega^{n-1}$ = all n roots cyclically (regular n -gon on unit circle, angle $2\pi/n$ apart); primitive = generator; n distinct roots of $z^n - 1 = 0$.
Primitive: ω_k^n generates all $\iff \gcd(k, n) = 1$; e.g. $n=8$: $k=1, 3, 5, 7$. **Why:** $\text{ord}(\omega_k^n) = n / \gcd(k, n)$ — shared factors shortcut the cycle (e.g. $\omega_8^8 = \omega_4$, order 4). **Gen test:** $g^d \neq 1$ for all proper divisors $d|n$ (Z13: $2^1, 2^2, 2^3, 2^4, 2^6 = 2, 4, 8, 3, 12 \neq 1$).
Cyclic: $C_n^* = \langle \omega_n \rangle$; reduce first: $\omega_8^2 = \omega_4 = i$.

Root: $\omega_n = e^{2\pi i/n}$.
Relations (n even; last two $\forall n$): $\omega^n = 1$, $\omega^{n/2} = -1$, $\omega^2 = \omega_{n/2}$, $\omega^{-1} = \bar{\omega}$, $\omega^{k+n/2} = -\omega^k$, $\omega^{2k} = \omega_{n/2}^k$, $\omega^{n-k} = \bar{\omega}^k$.

Worked $\omega_8 = (1+i)/\sqrt{2}$: $\omega_8^4 = -1$, $\omega_8^8 = 1$, $\omega_8^{-5} = \omega_8^{-5+8} = \omega_8^3 = (-1+i)/\sqrt{2}$;
 $\omega_{16}^4 = e^{8\pi i/16} = i = \omega_4$. $\sqrt{i} = (1+i)/\sqrt{2}$ (**Check** $((1+i)/\sqrt{2})^2 = i$); $\sqrt{-i} = (-1+i)/\sqrt{2}$.
Eval trick: reduce exponents mod n , use $\omega_2 = -1$.
Worked $p = x^4 - 3x^3 + x^2 - 2x - 1$ at $\omega_4 = i$: powers $\omega_4^2 = -1$, $\omega_4^3 = -i$, $\omega_4^4 = 1$.
Then $1 - 3(-i) + (-1) - 2i - 1 = 1 + 3i - 1 - 2i - 1 = -1 + i$.
Lecture: $3\omega_4^3 - 2\omega_4^2 + 2\omega_4^4 = 3(-i) + 2 + 2 \cdot 1 = 5 - 3i$.

5 — Iso $C_n^* \cong \mathbb{Z}_n^+$

Map $\omega_k^n \leftrightarrow [k]_n$; $\times \leftrightarrow +$:
 $\omega^k \omega^l = \omega^{k+l} \leftrightarrow [k] + [l] = [k+l]$. Gen $\leftrightarrow$ gen. **Iso:** bijection preserving op (C_4^* table $\equiv \mathbb{Z}_4^+$ table).
Subgroup: subset + closed under op.
Worked $C_4 \leftrightarrow \mathbb{Z}_4$: $[0] \leftrightarrow 1$, $[1] \leftrightarrow i$, $[2] \leftrightarrow -1$, $[3] \leftrightarrow -i$;
 $[1] + [2] = [3] \leftrightarrow i \cdot (-1) = -i$;
 $[3] + [2] = [5] = [1] \leftrightarrow (-i) \cdot (-1) = i$. Tables same shape $\Rightarrow$ iso.

No-iso test: $|G|$ must match + generator orders; e.g. $|C_2^*| = 2$ vs $|\mathbb{Z}_4^+| = 4 \Rightarrow$ no iso. **Fill C_5^* via iso:** $\omega_5^2, \omega_5^3 = \omega_5^2 = \omega_5^2 \leftrightarrow [3] + [4] = [7] = [2]$. Subgroups: $C_2^m \subset C_{2m+1}^*$ ($z^{2m} = 1 \Rightarrow (z^{2^m})^2 = 1$).

6 — Poly: FTA, Fields, Coeff/Value

Field: $+$, $-$, $\times$, $/$ (no $/0$); $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ yes; $\mathbb{Z}$ no (a/b can leave $\mathbb{Z}$); $\mathbb{N}$ no ($a-b$ can leave $\mathbb{N}$); $\mathbb{Z}_p$ (p prime) yes (every $\neq 0$ invertible). Polys over field K (usually $\mathbb{R}/\mathbb{C}$); some fields give no roots. Alg closed: every $\deg \geq 1$ has a root; $\mathbb{C}$ is (FTA); $\mathbb{R}$ is **not** ($x^2+1=0$ no real root).

FTA: $\deg n \Rightarrow$ **exactly n roots in $\mathbb{C}$** counting mult: $f = a_n \prod (x - x_i)$.

Worked $x^4 + 16 = 0$: no real roots; 4 in $\mathbb{C}$; two: $\sqrt{2} + i\sqrt{2}$, $-\sqrt{2} + i\sqrt{2}$ (**Check** $(\sqrt{2} + i\sqrt{2})^2 = 4i$, $^2 = -16$). All four: $\pm \sqrt{2} \pm i\sqrt{2}$ (signs independent); $|x|=2$, angles $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$.
4th roots of -1 : $-1 = e^{i\pi}$: angles $(\pi+2k\pi)/4 = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4 \Rightarrow (\pm 1 \pm i)/\sqrt{2}$ = odd ω_8 powers (all primitive).
Proof (no real roots): for real x , $x^4 \geq 0$ (even power), so $x^4 + 16 \geq 16 > 0$ — never 0. I.e. $x^4 = -16$ impossible over $\mathbb{R}$ (LHS ≥ 0). Hence all 4 FTA roots are nonreal. Factor thm:
 $f(a) = 0 \iff (x-a) \mid f$; $(x-1)^2 = x^2 - 2x + 1$ (multiple root); $x^2 + 1 = 0$: roots $\pm i$ (none real).
Note: 5 prime in $\mathbb{Z}$ but $5 = (2-i)(2+i)$ in $\mathbb{Z}[i]$.

Coeff ($a_0..a_n$); **value:** $\deg \leq n$ unique from $n+1$ pts (Vandermonde det $\neq 0$). **Product $\deg n+m$ needs $n+m+1$ pts;** pad zeros $(4+5x \rightarrow 4+5x+0x^2)$. Pointwise $h(x_i) = f(x_i)g(x_i)$, $O(m)$; schoolbook $O(n^2)$.
Worked $f=1-x, g=2+x$ at $-2, 0, 1$:
 $f: 1 - (-2) = 3, 1, 0$; $g: 2 + (-2) = 0, 2, 3$;
 $h: 0, 2, 0 \Rightarrow h = 2 - x - x^2$ (**Check** $2+2-4=0, 2, 2-1-1=0$).
Sketch recipe: mark roots $\rightarrow$ lead-coeff sign for ends $\rightarrow y$ -intercept $\rightarrow$ join; graphical mult: sample ≥ 6 x 's, multiply y 's, plot.

Point-count recipe: single $\deg-n \rightarrow n+1$ pts; product $\deg n+m \rightarrow n+m+1$; FFT $\rightarrow$ next pow2 $\geq$ that.

7 — FFT/IFFT + Costs

Even/odd: $p(x) = p_e(x^2) + x p_o(x^2)$. **Worked** $1-2x+3x^2-4x^3+5x^4-6x^5+7x^6-8x^7$:
 $p_e(y) = 1+3y+5y^2+7y^3$, $p_o(y) = -2-4y-6y^2-8y^3$;
depth of recursion tree = 3 ($8=2^3$).

Butterfly ($k=0..n/2-1$):
 $p(\omega^k) = p_e(\omega^{2k}) + \omega^k p_o(\omega^{2k})$;
 $p(\omega^{k+n/2}) = p_e(\omega^{2k}) - \omega^k p_o(\omega^{2k})$. Needs $n=2^t$.

Butterfly $n=4$ ($\omega=i$):

$k=0$: $p(1) = p_e(1) + p_o(1)$
 $p(-1) = p_e(1) - p_o(1)$
 $k=1$: $p(i) = p_e(-1) + i p_o(-1)$
 $p(-i) = p_e(-1) - i p_o(-1)$

NFFT $\geq 2n-1$, smallest pow2 ($4+4$ coeffs $\deg 3$: product $\deg 6$, 7 coeffs $\Rightarrow$ **N=8**). Pipeline: $\text{coeff} \rightarrow \text{FFT} \rightarrow \text{vals} \rightarrow \text{pointwise } \times \rightarrow \text{IFFT} \rightarrow \text{coeff}$. Zero-pad; extra points OK if same poly. **Idea:** one (p_e, p_o) pair gives two values (halves work).

IFFT: $a_j = \frac{1}{n} \sum_{k=0}^{n-1} y_k (\omega^{-j})^k$, $j=0..n-1$.

Costs: FFT/IFFT $O(n \log n)$ (depth $\log n \times O(n)/\text{level}$); pointwise $O(n)$; total $O(n \log n)$.

Worked FFT $q = 1 - x^2 + 4x^3$ ($N=4$, $\omega_4=i$):
 $q(1)=4$; $q(i)=2-4i$; $q(-1)=-4$; $q(-i)=2+4i$.
Butterfly check: $p_e(y) = 1-y$, $p_o(y) = 4y$;
 $q(i) = p_e(-1) + i p_o(-1) = 2 + i(-4) = 2 - 4i$.

Butterfly table: $k=0$: $p_e(1)=0$, $p_o(1)=4 \Rightarrow q(1)=0+4=4$, $q(-1)=0-4=-4$.
 $k=1$: $p_e(-1)=2$, $p_o(-1)=-4 \Rightarrow q(i)=2+i(-4)=2-4i$, $q(-i)=2-i(-4)=2+4i$.

Full pipeline $f=1-x, g=2+x$ ($N=4$):

$F: 0, 1-i, 2, 1+i$; $G: 3, 2+i, 1, 2-i$;

$H = F \cdot G: 0, 3-i, 2, 3+i$.

IFFT: $a_0 = (0+3-i+2+3+i)/4 = 8/4 = 2$;

$a_1 = (-3i-1-2+3i-1)/4 = -1$;

$a_2 = -1 \Rightarrow h = 2 - x - x^2$ **Check OK**.

Worked IFFT $y = (-1, i, 1, 2)$: $(\omega^{-j})^k$ ($k=0..3$, $\omega=i$):

$j=0$: $1, 1, 1, 1$; $j=1$: $1, -i, -1, i$; $j=2$: $1, -1, 1, -1$;

$j=3$: $1, i, -1, -i$.

$a_0 = (y_0 + y_1 + y_2 + y_3)/4$

$= (-1 + i + 1 + 2)/4 = (2 + i)/4 = 0.5 + 0.25i$.

$a_1 = (y_0 + y_1(-i) + y_2(-1) + y_3(i))/4$

$= (-1 + 1 - 1 + 2i)/4 = (-1 + 2i)/4 = -0.25 + 0.5i$

($i \cdot (-i) = 1$).

$a_2 = (y_0 + y_1(-1) + y_2 + y_3(-1))/4$

$= (-1 - i + 1 - 2)/4 = (-2 - i)/4 = -0.5 - 0.25i$.

$a_3 = (y_0 + y_1(i) + y_2(-1) + y_3(-i))/4$

$= (-1 - 1 - 1 - 2i)/4 = (-3 - 2i)/4 = -0.75 - 0.5i$

($i \cdot i = -1$).

Check $p(i) = a_0 + a_1 i - a_2 - a_3 i = i = y_1$.

Write p : $p(x) = (0.5 + 0.25i) + (-0.25 + 0.5i)x$

$+ (-0.5 - 0.25i)x^2 + (-0.75 - 0.5i)x^3$.

8 — Int Mult, GCD, Systems

Int as poly base B : $325 \rightarrow 5+2x+3x^2$ ($p(10)=325$). Steps: polys $\rightarrow$ FFT $\rightarrow$ pointwise $\times \rightarrow$ IFFT $\rightarrow$ carry to $[0, B-1] \rightarrow$ eval B . **Carry:** $[2, 15, 7] \rightarrow [2, 5, 8] = 852$ (raw $2+150+700=852$;

notes typo 825). $5678 \rightarrow 8+7x+6x^2+5x^3$;
 $q(10)=5678, q(2)=86$. d decimals $\rightarrow [d/3]$
base-1000 digits: 120!: $199 \rightarrow 67$ ($66\cdot3=198<199$).
Idea: every digit must land in $[0, B-1]$ (carry the rest).
GCD monic (lead 1; unique up to $\times \text{const}$):
 $4x^2+12x-7 \rightarrow x^2+3x-7/4$; $\gcd(x^2-1, x-1)=x-1$
 $((x-1)(x+1))$. Euclid via synthetic division.
Worked $f=x^4-8x^3+17x^2+2x-24$,
 $g=x^3-6x^2+11x-6$:
Step 1 (divide): $f-(x-2)g=-6x^2+30x-36=-6h$,
 $h=x^2-5x+6$.
Step 2: $g \bmod h$ is 0 $\Rightarrow \gcd(f, g)=h=x^2-5x+6$.
Synthetic $(x^3-6x^2+11x-6)/(x-1)$: bring 1; $\times 1$:
 $1 \rightarrow -5$; $\times 1$: $-5 \rightarrow 6$; $\times 1$: $6 \rightarrow 0$.
Quot x^2-5x+6 , rem 0 (so $x=1$ root).
Overdetermined: $h=\gcd$; $\deg h=0 \Rightarrow$
inconsistent ($x-1=0, x+1=0$); $\deg \geq 1 \Rightarrow$ common
roots = roots of h . Mini: $x^2-1=0$ (± 1) + $x-1=0$
(1) $\Rightarrow$ common $x=1$.
Worked: $x^2-11x+30=0$: $25-55+30=0$,
 $36-66+30=0 \Rightarrow 5, 6$.
 $x^3-10x^2+27x-18=0$: test 6:
 $216-360+162-18=0$; roots 1, 3, 6 $\Rightarrow$ common **$x=6$** .
Resultant: det condition for common root
(eliminates a variable). 2×2 :
 $\text{Res}(x-1, x+1)=(1)(1)-(-1)(1)=2 \neq 0 \Rightarrow$ no
common root.

9 _____Worked Solutions (Step-by-Step) _____Extra Q&A (Quiz/HW Style)

FFT size Qs: p with n coeffs needs n points
forth+back; FFT needs pow2 so pad.
 $n-1=16 \Rightarrow n=17$ (not pow2) $\Rightarrow$ **cannot use
directly**, pad to 32. $(p(x))^2$: $\deg 2(n-1)$, needs
 $2n-1$ pts $\rightarrow$ next pow2.
IFFT $n=2, y=(1, 1+i)$: $x_0=1=\omega_2^0, x_1=-1=\omega_2^1$;
 $a_0=(y_0+y_1)/2=(2+i)/2=1+0.5i$;
 $a_1=(y_0-y_1)/2=-0.5i$. **Check:** $p(1)=1$,
 $p(-1)=1+i$. Gen a_2 formula ($n=4$):
 $a_2=(y_0-y_1+y_2-y_3)/4$ ($\omega^{-2k}=1, -1, 1, -1$). **Idea:**
IFFT = FFT with ω^{-1} , then $/n$ (**trap:** forgetting
 $/n$ or the -1).
 C_8^* products ($z_1=1, z_2=-1, z_5=(1+i)/\sqrt{2}$,
 $z_6=(-1+i)/\sqrt{2}, z_7=(1-i)/\sqrt{2}, z_8=(-1-i)/\sqrt{2}$):
 $z_2z_5=(-1-i)/\sqrt{2}=z_8$; $z_5z_7=1=z_1$; $z_6z_8=1=z_1$.
Inverse of $(-1-i)/\sqrt{2}$: $|z|^2=1$, inv
 $=\bar{z}=(-1+i)/\sqrt{2}=z_6$.
Group tests: $2a+7bi$ ($a, b \in \mathbb{Z}$) additive: closed,
 $e=0$, inv $-z \Rightarrow$ **group**. $3a+3bi, ab \neq 0$
multiplicative: $1=3a+3bi \Rightarrow a=1/3 \notin \mathbb{Z} \Rightarrow$ **not** (no
identity).
FFT $n=32$: depth 5 ($2^5=32$); butterfly pairs
 $(\omega^0, \omega^{16}), (\omega^1, \omega^{17})$; even/odd parts $\deg 15$, split
again $\rightarrow 4$ polys ($p_{ee}, p_{eo}, p_{oe}, p_{oo}$). C_n^+ = cyclic
additive order n ($\cong \mathbb{Z}_n^+$), gens $[k]$ with $\gcd(k, n)=1$;
chain $C_{2^m}^* \subset C_{2^{m+1}}^*$.
Formulas: $p(\omega^{\frac{n}{2}})=p_e(1)+p_o(1)$,
 $p(\omega^{16})=p_e(1)-p_o(1)$; $p(\omega^1)=p_e(\omega^2)+\omega p_o(\omega^2)$,
 $p(\omega^{17})=p_e(\omega^2)-\omega p_o(\omega^2)$.
Inconsistent check: $\gcd(x-1, x+1)=1$ ($\deg 0$) $\Rightarrow$
inconsistent $((x+1)-(x-1)=2 \neq 0)$.

_____Mod + Groups/FLT Drills

Reduce first: $[125]_{17}+[241]_{17}=[366]_{17}$;
 $366-21\cdot 17=366-357=9$. $[391]_{21}\cdot [457]_{21}$:
 $391-18\cdot 21=13, 457-21\cdot 21=16$;
 $13\cdot 16=208-9\cdot 21=19$. $[94]_7: 94-13\cdot 7=94-91=3$.
 $[157]_{13}\cdot [249]_{13}\cdot [83]_{13}: 157-12\cdot 13=1$,
 $249-18\cdot 13=249-234=15; 15-13=2; 1+2=3$;
 $83-6\cdot 13=5; 3\cdot 5=15-13=2$.

HW08: $[123]_{17}+[245]_{17}=11$ ($368-357$).
 $[391]_{23}\cdot [457]_{23}=0$ ($391=17\cdot 23$).
 $([156]_{11}+[247]_{11})\cdot [83]_{11}: 2+5=7, \times 6=42-33=9$.
 $[312]_{15}\cdot ([184]_{15}+[257]_{15}): 12\cdot (4+2)=72-60=12$.
Find x :
 $143 \equiv x \pmod{13}: 143=11\cdot 13 \Rightarrow 0$
 $12345 \bmod 11: 11\cdot 1122=12342 \Rightarrow 3$
 $54321 \bmod 7: 7\cdot 7760=54320 \Rightarrow 1$
 $147 \bmod 13: 13\cdot 11=143 \Rightarrow 4$
Find n : $24 \equiv 0 \pmod{n} \Rightarrow n|24$:
1, 2, 3, 4, 6, 8, 12, 24. $15 \equiv 1 \Rightarrow n|14$: 1, 2, 7, 14.
 $124 \equiv 4 \Rightarrow n|120, n>4$:
5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120. $18 \equiv 0$,
 $0<n \leq 18$: 1, 2, 3, 6, 9, 18.
Classes: mod 7: 7 classes $[0]..[6]$.
 $[2]_5=\{\dots, -8, -3, 2, 7, 12, \dots\}$; $[4]_7=\{\dots, -3, 4, 11, \dots\}$;
 $[-3]_8=[5]_8=\{\dots, -3, 5, 13, \dots\}$. $\{1, -1\} \subset C_4^*$
subgroup (closed, $e=1$, self-inv).
Notation $\leftrightarrow$: $[8]_{11}+[9]_{11}=[6]_{11} \iff 8+9 \equiv 6$
(mod 11) ($17-11=6$); $[4]_7\cdot [5]_7=[6]_7 \iff 4\cdot 5 \equiv 6$
(mod 14); $17+9 \equiv 2$
(mod 8) $\iff [17]_8+[9]_8=[2]_8$ ($26-24=2$); $6\cdot 8 \equiv 3$
(mod 5) $\iff [6]_5\cdot [8]_5=[3]_5$ ($48-45=3$).
Groups: $(\mathbb{Q}, +)$ group; $(\mathbb{Q}, \cdot)$ **not** (0 no inverse).
 $(\mathbb{Z}_4, +)$ group ($e=[0]$, inv $[-a]$). $\mathbb{Z}_4^*=\{1, 2, 3\}$ **not**
($2<2=0$). Mult groups on $\mathbb{Z}_n$: $n=3, 5, 13, 17$ yes
($|G|=2, 4, 12, 16$); 10, 15 no. $\mathbb{Z}_5^*$ gens $\{2, 3\}$
(ord=4).
Possible orders: $2 \rightarrow \{1, 2\}$; $4 \rightarrow \{1, 2, 4\}$;
 $12 \rightarrow \{1, 2, 3, 4, 6, 12\}$; $16 \rightarrow \{1, 2, 4, 8, 16\}$.
 $|\mathbb{Z}_{24}^*|=\varphi(24)=8$ (odd, not mult of 3:
1, 5, 7, 11, 13, 17, 19, 23); x odd (e.g. 3, 5, 7) never
gives $[a]^x=[1]$ for $a=2\cdot 23$ (needs $2|x$; $[5]^2=[1]$ so
every even x works).
Gen of $\mathbb{Z}_{13}^*$: $|G|=12$, orders $\in \{1, 2, 3, 4, 6, 12\}$;
 $2^2=4, 2^3=8, 2^4=3, 2^6=12 \neq 1 \Rightarrow \text{ord}(2)=12 \Rightarrow$
generator.
FLT big powers: $19^6 \bmod 7$ OK ($7|19$); 21^6 **no**
(gcd=7); $11^8 \bmod 9$ **no** (9 composite). 19^{68} :
 $19 \equiv 5, 68=6\cdot 11+2$ ($66+2$), $5^2=25 \equiv 4$.
 $271^{24} \bmod 11$: $271-24\cdot 11=271-264=7$,
 $24=10\cdot 2+4, 7^4=2401-218\cdot 11=2401-2398=3$.
 $290^{293} \bmod 23$: $290-12\cdot 23=290-276=14$,
 $293=22\cdot 13+7$ ($286+7$), $14^7 \equiv 19$. No-FLT:
 $[16^4]_7=[2]$; $[10^{100}]_9=[1]$; $[23^2]_9=[7]$ ($23 \equiv 5$,
 $25-18$); $[42]_{12}=[6]$.
More Euler: $5^{100} \bmod 13$: $\varphi=12$; $100=12\cdot 8+4$;
 $5^4=625-48\cdot 13=625-624=1$. $2^{1000} \bmod 7$:
 $1000=6\cdot 166+4$; $2^4=16 \equiv 2$. $\varphi(8)=8-4=4$;
 $\varphi(24)=\varphi(8)\varphi(3)=4\cdot 2=8$;
 $\varphi(100)=\varphi(4)\varphi(25)=2\cdot 20=40$. **Trap:**
 $\varphi(4)\varphi(2)=2\cdot 1=2 \neq 4=\varphi(8)$ (gcd(4, 2) $\neq 1$).
More cipher: HI, $s=20$: $7+20=27-26=1=B$,
 $8+20=28-26=2=C \Rightarrow$ **BC**. Power $m=2, p=29$,
 $b=5$: $2^5=32-29=3$. **Trap:** $2^4 \bmod 8=0 \neq 1$ though
 $\varphi(8)=4$ (gcd $\neq 1$).
Mod x^3-1 (Quiz4 Q6): $x^3 \equiv 1$:
 $[2x^5-9x^3+5x+9]=2x^2-9+5x+9=[2x^2+5x]$.
Square after reducing:
 $[(2x^2+5x)^2]=[4x^4+20x^3+25x^2]$; $x^4 \equiv x$,
 $x^3 \equiv 1 \Rightarrow [25x^2+4x+20]$.

_____Complex + FFT Drills

Sums (plot tip-to-tail, check):
 $(-2+4i)+(-3-7i)=-5-3i$
 $(-2-i)+(2+i)=0$
 $(1-2i)+(1+2i)=2$
 $|z|^2=zz\bar{z}$: $(-2-3i): 4+9=13; 4i: 16$;
 $\bar{z}=4i \Rightarrow z=-4i, |z|^2=16$.
Div: $(2+6i)/2=1+3i$.
 $(3+4i)/(1-6i)=(3+4i)(1+6i)/37=(-21+22i)/37$.
 $\approx -0.5676+0.5946i$ **Check** ($37\cdot (-0.5676)=-21$,

$37\cdot 0.5946=22$). **vs dot:** $z_1z_2=(ac-bd)+(ad+bc)i$
vs $u\cdot v=ac+bd$; $|z_1z_2|^2=(a^2+b^2)(c^2+d^2)$ vs
 $(u\cdot v)^2=(ac+bd)^2$; coincide $\iff ad=bc$.
E.g. $z_1=1+2i, z_2=3+4i$:
 $|z_1z_2|^2=(1+4)(9+16)=125$ vs $(3+8)^2=121$
($1\cdot 4 \neq 2\cdot 3$).
Even/odd twice of $p_e=1+3y+5y^2+7y^3$,
 $p_o=-2-4y-6y^2-8y^3$:
 $p_{ee}(z)=1+5z, p_{eo}(z)=3+7z$
 $p_{oe}(z)=-2-6z, p_{oo}(z)=-4-8z$
Big-O tree (n leaves, balanced binary): depth
 $\log_2 n$; bottom-only work $\Rightarrow O(n)$; $O(n)$ per level
 $\Rightarrow O(n \log n)$.
 C_5^* row via iso:
 $\omega_5^4\cdot \omega_5^4=\omega_5^8=\omega_5^3 \leftrightarrow [4]+[4]=[8]=[3]$. Gen $\omega_5 \leftrightarrow [1]_5$
powers fill table.

_____Tables + FFT Drills

Mod-8 $\times$ table (nonzero rows): Rule: row a
has 1 $\iff a$ invertible; row a hits 0 $\Rightarrow$ zero
divisor. Units = odd $\{1, 3, 5, 7\}$ (gcd=1); $1^{-1}=1$,
 $3^{-1}=3$ ($9-8=1$), $5^{-1}=5$ ($25-24=1$), $7^{-1}=7$
($49-48=1$). Even rows lack 1. **Zero divisors:**
 $2\cdot 4=8 \equiv 0, 6\cdot 4=24 \equiv 0$ ($24-24$).
FFT $f=2+3x+5x^3$ ($N=4, \omega_4=i$; $p_e(y)=2$,
 $p_o(y)=3+5y$): $f(1)=2+3+5=10$;
 $f(i)=2+3i+5(-i)=2-2i$; $f(-1)=2-3-5=-6$;
 $f(-i)=2-3i+5i=2+2i$.
Butterfly table (f): $k=0$: $p_e(1)=2$,
 $p_o(1)=8 \Rightarrow f(1)=2+8=10, f(-1)=2-8=-6$.
 $k=1$: $p_e(-1)=2$,
 $p_o(-1)=-2 \Rightarrow f(i)=2+i(-2)=2-2i$,
 $f(-i)=2-i(-2)=2+2i$.
DeMoivre power:
 $((1+i)/\sqrt{2})^8=(e^{i\pi/4})^8=e^{2\pi i}=1$ (angle $\times 8$).
Decrypt proof: $m \equiv$
 $c^d=(m^b)^d=m^{bd}=m^{1+k(p-1)}=m\cdot (m^{p-1})^k \equiv m$
(FLT, p prime, $p|m$).

_____Tables + Step-by-Step Recoveries

Mod-8 $\times$ table (read inverses/zero divisors; odd
rows permute, even rows hit 0):

$\times$	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7
2	2	4	6	0	2	4	6
3	3	6	1	4	7	2	5
4	4	0	4	0	4	0	4
5	5	2	7	4	1	6	3
6	6	4	2	0	6	4	2
7	7	6	5	4	3	2	1

C_4^* vs $\mathbb{Z}_4^+$ (same shape $\Rightarrow$ iso $\omega_4^k \leftrightarrow [k]_4$):

*	1	i	-1	$-i$	+	0	1	2	3
1	1	i	-1	$-i$	0	0	1	2	3
i	i	-1	$-i$	1	1	1	2	3	0
-1	-1	$-i$	1	i	2	2	3	0	1
$-i$	$-i$	1	i	-1	3	3	0	1	2

C_5^* table via iso ($\omega=\omega_5; \omega^i\cdot \omega^j=\omega^{(i+j) \bmod 5}$):

*	1	ω	ω^2	ω^3	ω^4
1	1	ω	ω^2	ω^3	ω^4
ω	ω	ω^2	ω^3	ω^4	1
ω^2	ω^2	ω^3	ω^4	1	ω
ω^3	ω^3	ω^4	1	ω	ω^2
ω^4	ω^4	1	ω	ω^2	ω^3

**Recover $h=a+bx+cx^2$ from values (0, 2, 0) at
(-2, 0, 1):** $a-2b+4c=0, a=2, a+b+c=0$. **Step 1:**
 $a=2$. **Step 2:** $b+c=-2, -2b+4c=-2 \Rightarrow b-2c=1$.
Step 3: subtract: $3c=-3 \Rightarrow c=-1, b=-1$. **Box:**
 $h=2-x-x^2$. Check: $2+2-4=0, 2, 2-1-1=0$.
Read graph $\rightarrow$ number: point $(x, y) \rightarrow x+yi$;
length r , angle $\theta \rightarrow r(\cos \theta + i \sin \theta)$.